

**Prediction and Optimization of Wear Properties of
copper- graphite-TiC composite under lubricating sliding
condition by using Machine Learning**



**KAMLA NEHRU INSTITUTE OF TECHNOLOGY,
SULTANPUR (U.P.)**

Presented to -
**Head of Department
Mechanical Engineering**

Presented by
**Gaurav Kumar (22521)
Manas Srivastava (22528)
Sameer Kanaujia (22547)
Sudhanshu Shukla (22554)**

- 1. Introduction**
- 2. Literature Review**
- 3. Research Gap**
- 4. Objective and Scope**
- 5. Materials and Method**
- 6. Tribological Testing**
- 7. Literature Insights on Cu-Gr-TiC**
- 8. Challenges / Why ML-based Approach is Suitable**
- 9. Data and Feature**
- 10. Methodology**
- 11. Workflow of Machine Learning**
- 12. Exploratory Data Analysis**
- 13. Artificial Neural Network**
- 14. Random Forest (RF) Regression**
- 15. Support Vector Regression (SVR)**
- 16. K-Nearest Neighbors (KNN) Regression**
- 17. Comparative Analysis of Machine Learning Model**
- 18. Test Result**
- 19. Conclusions**
- 20. References**

INTRODUCTION

Tribology Focus: Study of friction, wear, and lubrication in moving surfaces .

Industrial Impact: High friction and wear cause material decay, energy loss, and system failure.

Material Choice: Copper-based composites offer great thermal conductivity and corrosion resistance.

The Solution: Adding Graphite (for self-lubrication) and Titanium Carbide (for hardness and strength).

Research Gap & AI Integration Traditional Limits: Manual lab testing is slow, expensive, and resource-heavy.

ML Advantage: Machine learning uncovers hidden data patterns to predict properties accurately.

Methodology & Objectives Four Tested Models: Support Vector Regression, K-Nearest Neighbors, Artificial Neural Network, and Random Forest.

Input Features: Applied load, sliding distance, and reinforcement composition.

Output Targets: Wear rate and coefficient of friction (COF).

Evaluation Metrics: MAE, MSE, RMSE, and R^2 score.

Core Goal: Create a reliable, data-driven framework to cut experimental costs and speed up material design.

LITERATURE REVIEW

S.No.	Paper	Work	Result
1	Ankit et al.	Prediction of Tribological performance of Cu-Gr-Tic composites based on response surface methodology and worn surface analysis.	TiC reinforcement lowers the composite wear rate but increases the coefficient of friction by 11%. Higher applied loads significantly increase both wear rate and friction.
2	Huifeng NING, Faqiang CHEN, Yunfeng SU, et al.	Modeling and Prediction of Tribological properties of Copper/aluminum-graphite self-lubricating composites using machine learning	Ensemble models, particularly LSBoost ($R^2=0.9219$) and XGBoost ($R^2=0.9158$), outperformed traditional methods in predicting composite friction coefficients, proving superior to bagging and KNN models in this analysis.
3	Ankit, Gaurav Gautam, Kamlesh K Singh and Sunil Mohan	Synergetic influence of TiCnp and graphite particles on tribological performance of Cu based composites prepared by flake powder metallurgy	Adding 4.5% TiC and 5% graphite to copper significantly boosts hardness and wear resistance while maintaining lubrication, making it ideal for bearings and electrical contacts.

Research Gap

- Several studies have investigated the tribological behavior and wear characteristics of Cu–TiC–Gr composites through experimental methods. Most studies focus primarily on experimental analysis and statistical interpretation of results.
- Limited research has been conducted on the application of Machine Learning techniques for predicting the wear behavior of Cu–TiC–Gr composites. Existing studies rarely compare the performance of different machine learning models such as Artificial Neural Networks (ANN), K-Nearest Neighbors (KNN), Support Vector Regression (SVR), and Random Forest (RF) for accurate wear prediction. Therefore, there is a need to develop a data-driven predictive framework that can estimate wear behavior efficiently and reduce the dependency on extensive experimental trials.
- The present work aims to bridge this gap by employing and comparing multiple machine learning algorithms to predict the wear rate of Cu–TiC–Gr composites using experimental data obtained from previous tribological studies.

Objectives & Scope

1. Develop predictive models for wear rate / friction behavior of a Cu-Graphite-TiC composite.
2. Compare multiple machine learning algorithms (e.g. ANN, Random Forest, KNN, SVR) to find the best performer.
3. Use existing experimental wear data to train & validate the models.
4. Provide an optimization / predictive tool to estimate wear rate — reducing need for extensive experiments.

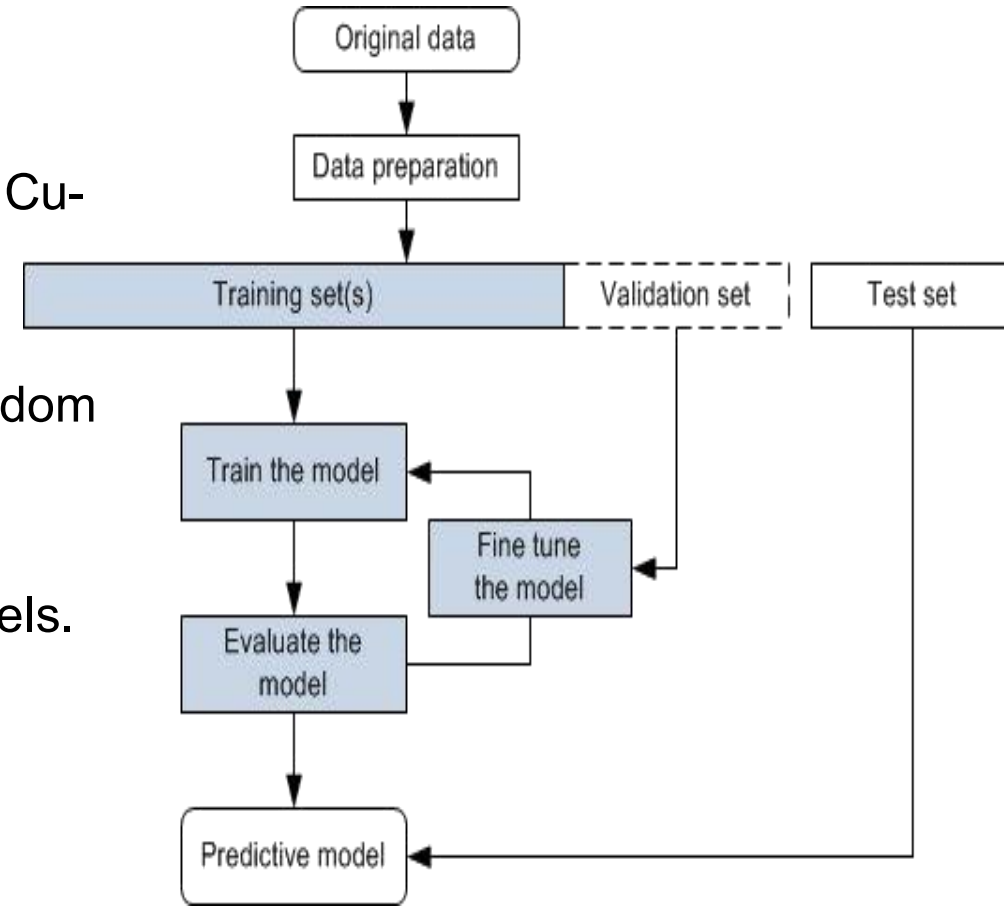


Fig.2.5. Friction and Wear Degradation Cycle

Materials And Methods

- ❖ **Raw materials-** Copper powder (94%), Graphite powder (5%), and TiC powders (1%).
- ❖ **Procured from-** Molychem India, CDH India and Sigma- Aldrich, respectively.

Table 1.1. Powders size

Materials	Average Particle Size
Cu	45µm
Gr	7-11µm
TiC	<200nm

Table 1.2. Nomenclature

S.NO.	COMPOSITIONS	NOMENCLATURE
1.	Pure Copper	Cu
2.	Copper – 5wt.% Graphite	Cu-Gr
3.	Copper - 5wt.% Graphite – 1.5wt.%TiC	T1.5
4.	Copper- 5wt.% Graphite-3.0wt.%TiC	T3.0
5.	Copper- 5wt.% Graphite-4.5wt.%TiC	T4.5

Materials And Methods

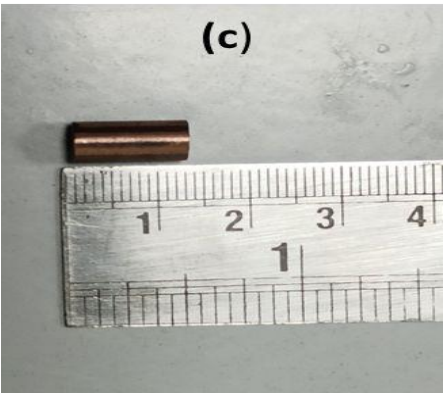
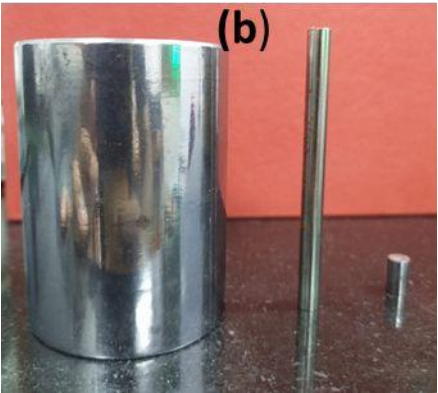
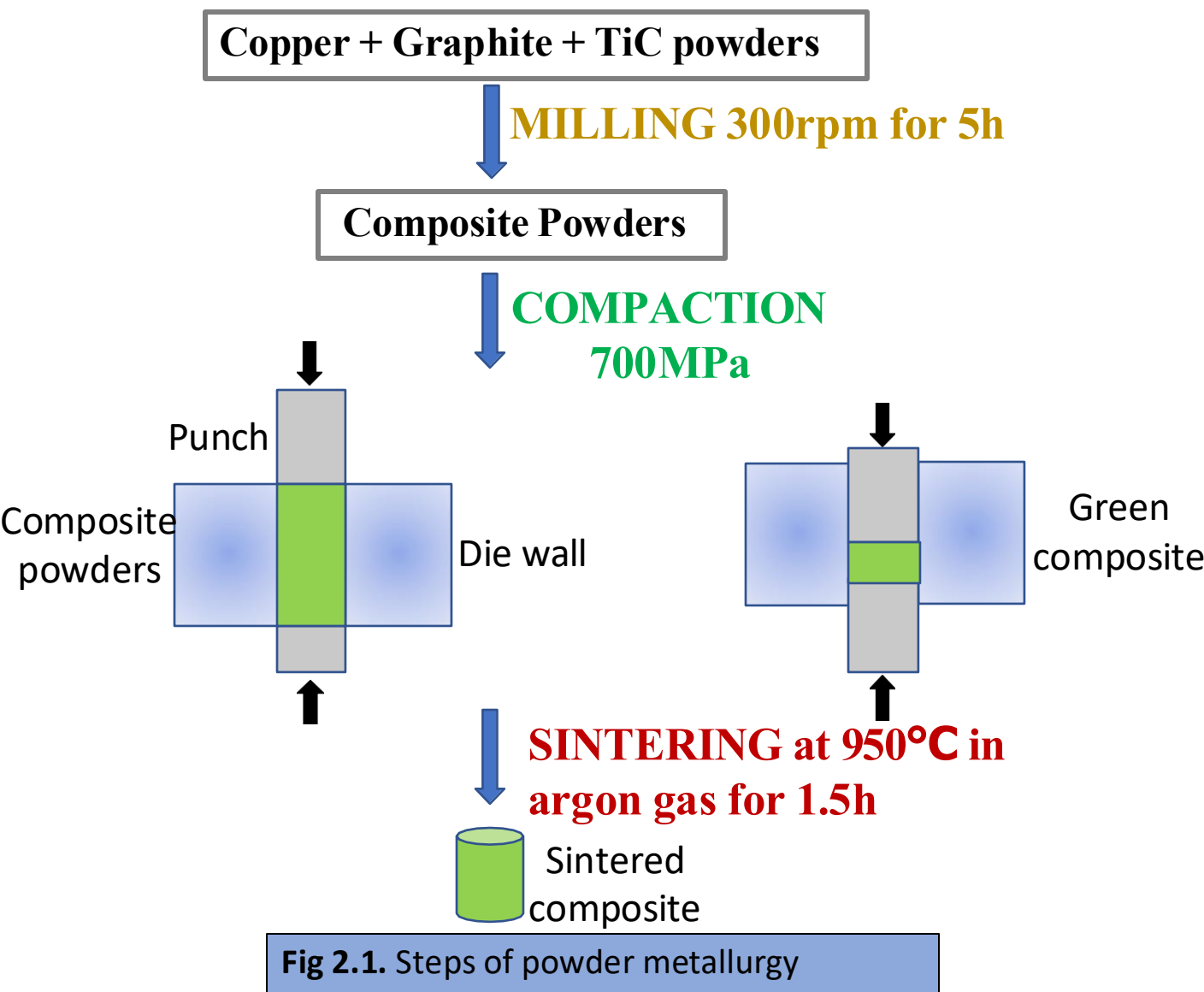


Fig 2.2.(a) PM400 milling machine, (b) Compaction die and punch and (c) Sintered sample

Tribological Testing

- ❖ Dry and Lubricating sliding tests were performed on samples using Pin On Disc tribometer. Weight loss was measured manually using mettler toledo weighing balance with least count of 0.1mg. The wear is calculated in terms of wear rate (mm^3/m).

Table 1.3. Parameters used for tribological test according to ASTM standard ASTM G99-17

Condition	Load (N)	Sliding Distance (m)	Sliding Velocity (m/s)	Testing condition	Pin Sample size (mm)	Counter disc
Lubrication	30-70	4000-8000	1.5	Room Temp.	6×12	EN31 Steel



Fig. 2.3. Pin on Disc tribometer setup

Literature Insights on Cu–Graphite-TiC/ Composites

- In metal-matrix composites like copper–graphite, graphite acts as a solid lubricant: it reduces friction coefficient and can form a lubricating tribo-layer on sliding surfaces.
- However, wear behavior depends strongly on graphite content: below a critical threshold of graphite concentration, friction & wear decrease significantly, but above that threshold the behavior may worsen due to weakening of matrix properties or poor bonding.
- Wear mechanisms evolve with composition: e.g. at low graphite — adhesive wear; at higher graphite — abrasive wear or increased wear rate due to metal-matrix weakening.
- Hardness / microstructure (matrix bonding, particle distribution) strongly influences wear rate: as hardness / matrix integrity decreases (e.g. with too much graphite or poor particle bonding), wear rate may increase.

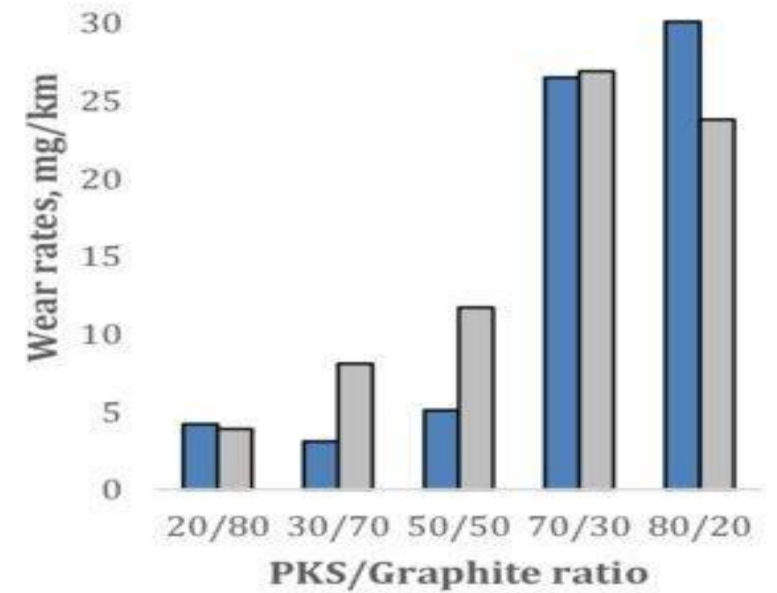


Fig.2.4. Wear Rate VS Graphite Ratio

Challenges / Why ML-based Approach is Suitable

- Multi-parameter dependence: wear depends on composition, load, speed, lubrication, material microstructure — complex, often non-linear relationships.
- Traditional analytical/theoretical models might not capture combined effects, especially for heterogeneous composites.
- Experimental testing for every combination is expensive and time-consuming.
- ML models (ANN, Random Forest, KNN, SVR) excel at capturing complex, non-linear relationships from data.
- Once trained and validated, model can quickly predict wear for new compositions/conditions — saving effort & cost.

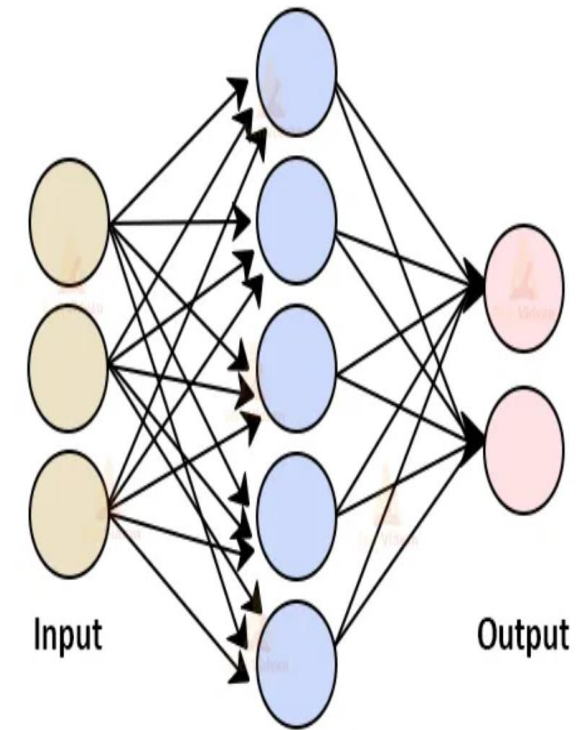


Fig.2.6. Input Output Cycle

Data & Features

- Input parameter (possible / required):
 - Composite composition (variation of TiC wt%).
 - Sliding conditions: load (normal load 30-70N), sliding speed(4000-8000m), sliding distance / duration, lubrication condition (dry / lubricated).
- Output / Predicted values:
 - Wear rate
 - Coefficient of friction
- Dataset status : experiment data compiled in “Wear Rate Information” / data files; used for model training.

Methodology

1. Data collection — gather experimental wear data (from own experiments or literature) for various compositions & conditions.
2. Data preprocessing — clean data, handle missing values, normalize / scale features, encode categorical variables (e.g. lubrication condition).
3. Exploratory Data Analysis (EDA) — visualize trends (wear vs composition, load, lubrication), check distributions, correlations.
4. Split dataset into training & test sets.
5. Model training: multiple ML models — e.g.
 - Artificial Neural Network (ANN) — captures non-linearities, complex interactions
 - Random Forest Regression — robust, handles noisy / small datasets
 - SVR (Support Vector Regression) -- It is a type of supervised learning algorithm used to predict continuous numerical values
 - KNN (K-Nearest Neighbors) -- It is one of the simplest and most widely used supervised learning algorithms.
6. Evaluate models — using metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), R^2 , comparison of predicted vs actual curves.
7. Model selection & optimization — tune hyperparameters, cross-validate, choose best model.

Workflow of Machine Learning

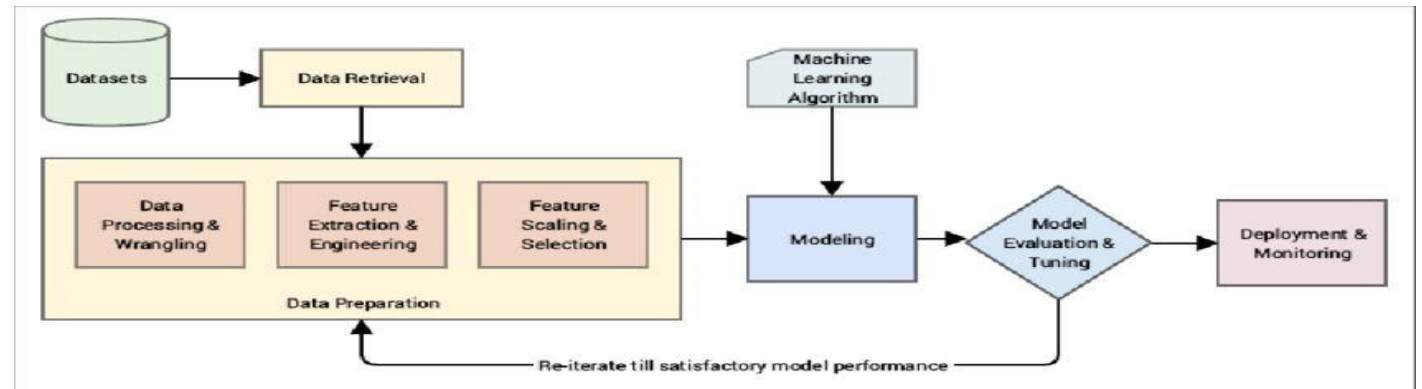
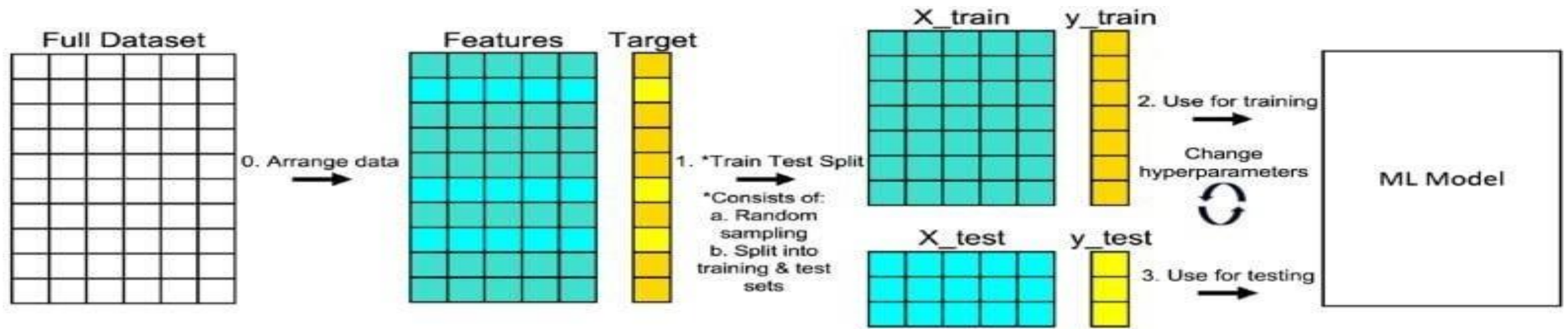


Fig.2.7. Structure of input and output chain Model

Exploratory Data Analysis

- Load vs. Wear Rate (+0.80): Strong positive correlation—the dominant tribological driving force.
- Load vs. COF (+0.77): Strong positive correlation — increased contact pressure for the real contact area.
- TiC Content vs. Wear Rate (−0.41): Moderate negative correlation — addition of hard TiC particles reduces material removal.
- TiC Content vs. COF (−0.56): Moderate-to-strong negative correlation — TiC reinforcement combined with graphite lubrication significantly reduces friction at the sliding interface.

The correlation analysis confirms that Load is the most influential parameter for both responses and that TiC content is the most effective design lever for improving tribological performance.

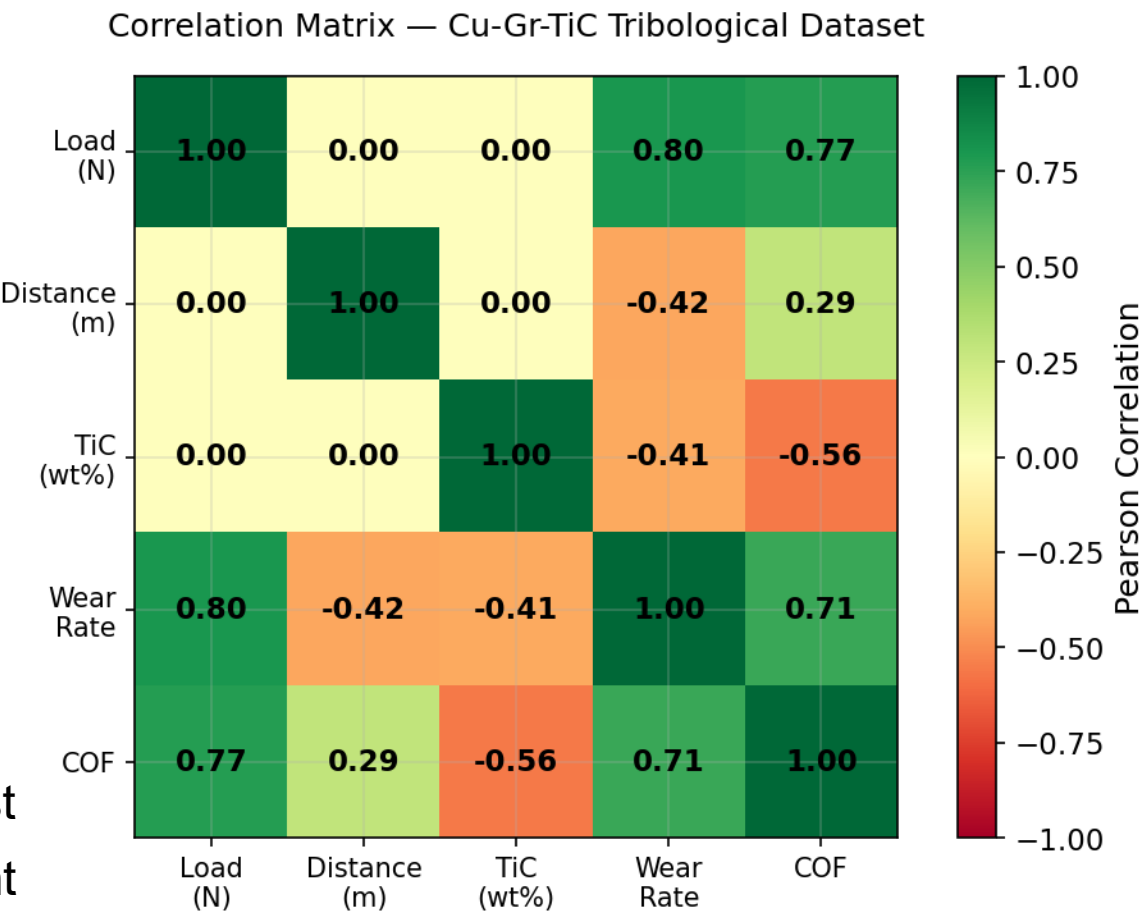


Fig.2.8. *Pearson Correlation Matrix*

Artificial Neural Network

- An Artificial Neural Network (ANN) is a supervised learning algorithm loosely inspired by the structure of biological neural networks in the human brain. The computation at each hidden layer is:

$$h_l = \text{ReLU}(W_l \cdot h_{l-1} + b_l)$$

- Figure 2.10. presents the predicted versus actual values for both outputs on the test set.

Table 1.4: ANN (MLP) Training Hyperparameters

Hyperparameter	Value	Description
Train-Validation Split	80:20 (of training set)	For early stopping criterion
Learning Rate	0.001	Initial step size for weight updates
Max Iterations	10,000	Upper bound on training epochs

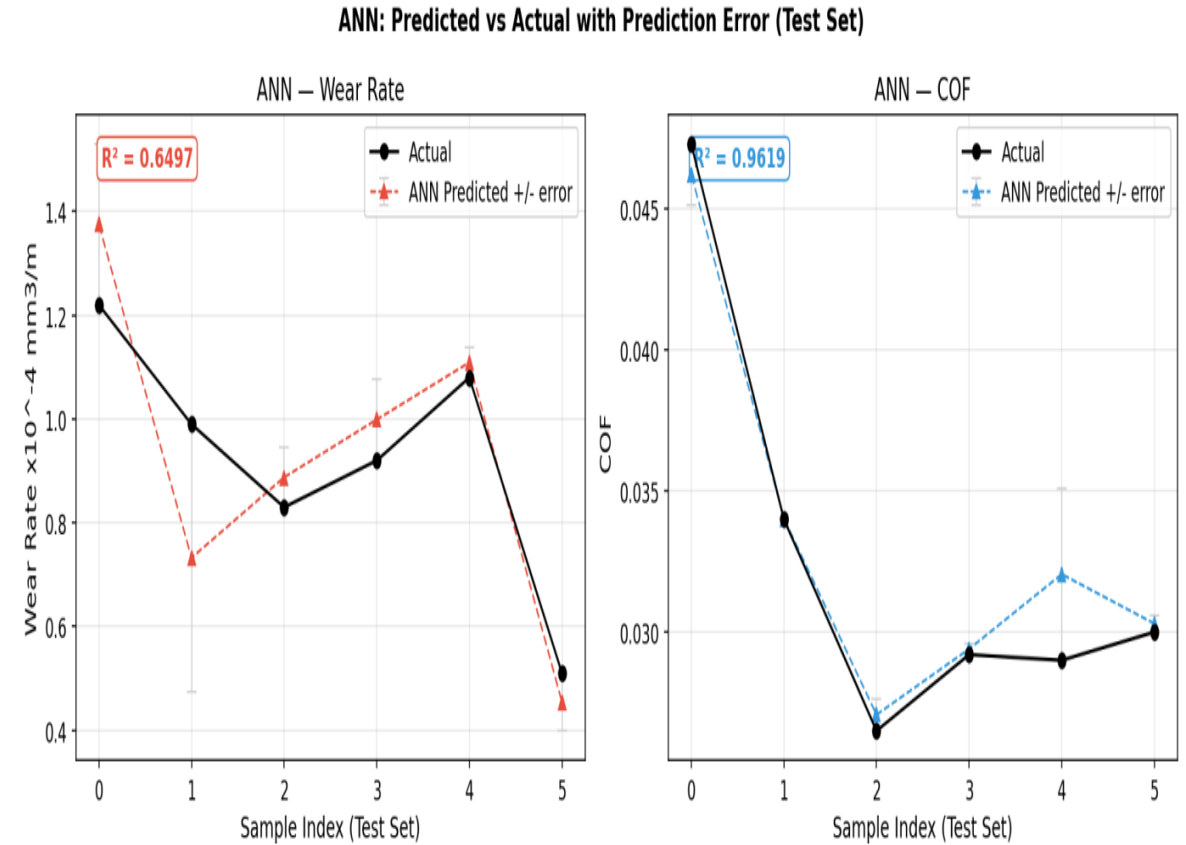


Fig.2.9. ANN Predicted vs. Actual Values

Random Forest (RF) Regression

- Random Forest is an ensemble learning method that constructs a large number of decision tree regressors during training and aggregates their predictions through averaging to produce the final output
- The final prediction is the mean of all individual tree predictions:

$$\hat{y}(x) = (1/T) \times \sum_{t=1}^T f_t(x)$$

where T is the total number of trees, $f_t(x)$ is the prediction of the t-th tree, and $\hat{y}(x)$ is the ensemble prediction

- Figure 2.11 compares the experimental versus RF-predicted values across. The predicted curves track the experimental trends closely for both Wear Rate and COF, with small but visible deviations at peaks and valleys

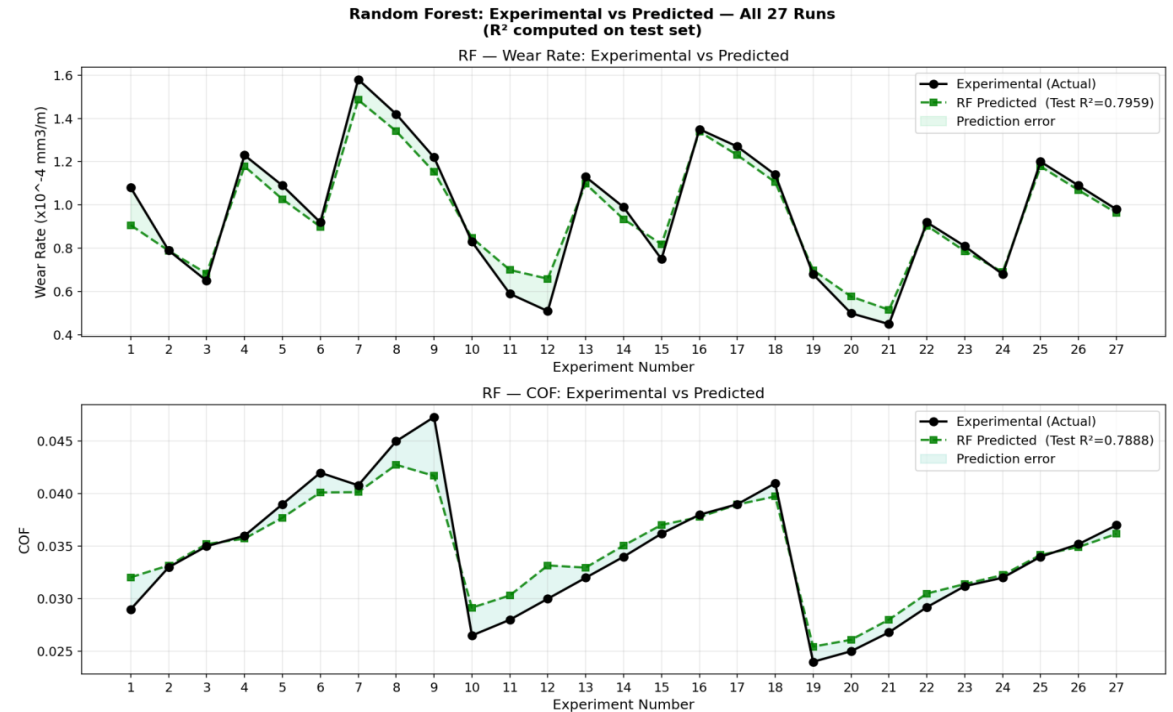


Fig.2.10. RF Predicted vs. Actual Values

Support Vector Regression (SVR)

- Support Vector Regression (SVR) extends the principles of Support Vector Machine (SVM) classification to continuous value prediction.
- Figure 2.12 shows the experimental versus SVR-predicted values across all 27 runs. The narrow shaded band between the predicted (red dashed) and actual (black solid) curves indicates consistently small prediction errors throughout the full experimental range.

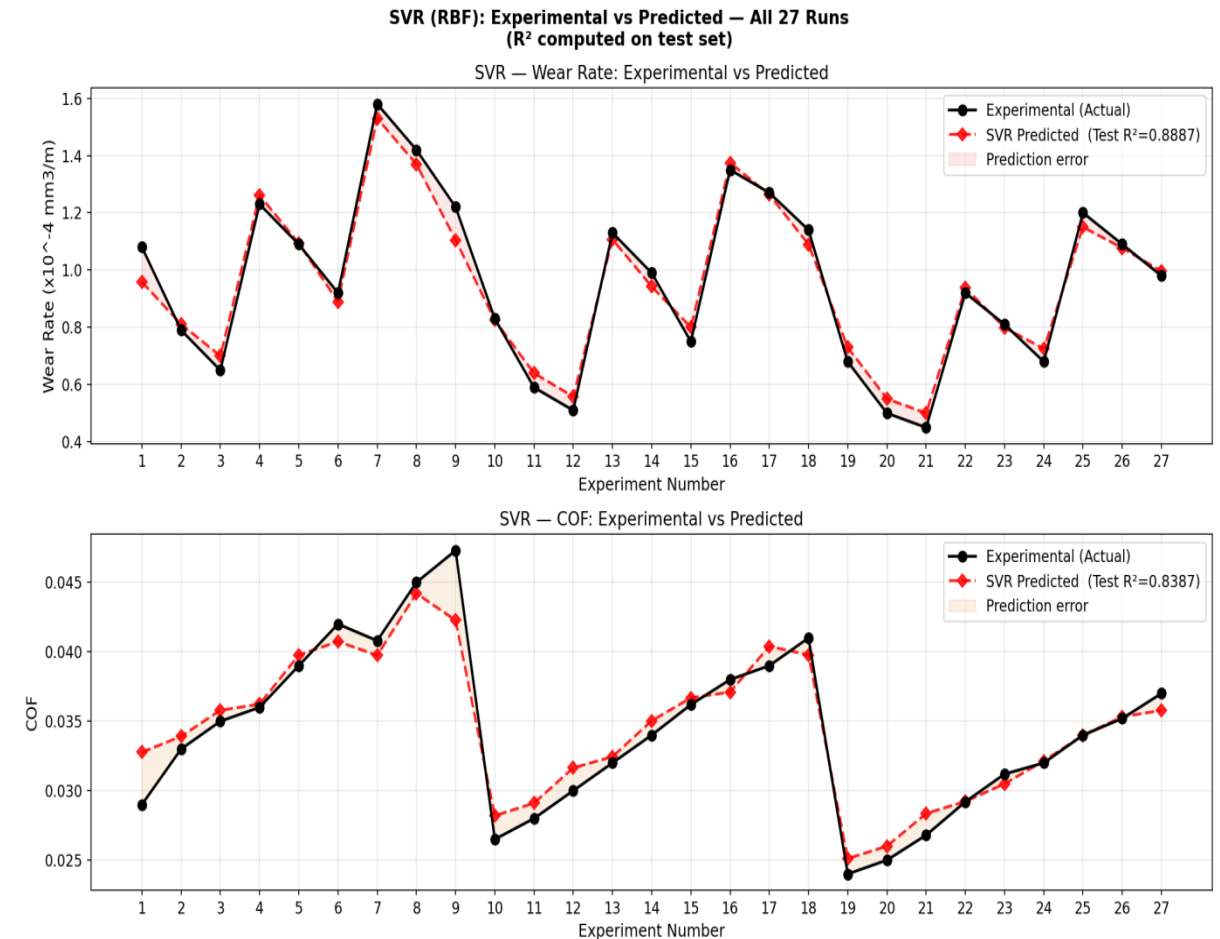


Fig.2.11. SVR Predicted vs. Actual Values

K-Nearest Neighbors (KNN) Regression

- K-Nearest Neighbors (KNN) is a non-parametric, instance-based (lazy) learning algorithm that makes predictions without explicitly constructing a model during training.

KNN algorithm proceeds as follows:

- Compute the Euclidean distance $d(x^*, x_i) = \sqrt{\sum_j (x_j^* - x_{ij})^2}$ between x^* and every training sample x_i .
- Select the K training samples with the smallest distances (the K nearest neighbors).
- Compute the prediction as the distance-weighted average: $\hat{y} = \sum_i w_i \cdot y_i / \sum_i w_i$, where weights $w_i = 1/d(x^*, x_i)$ give more influence to closer neighbors.
- Figure 2.13. shows the experimental versus KNN-predicted curves .

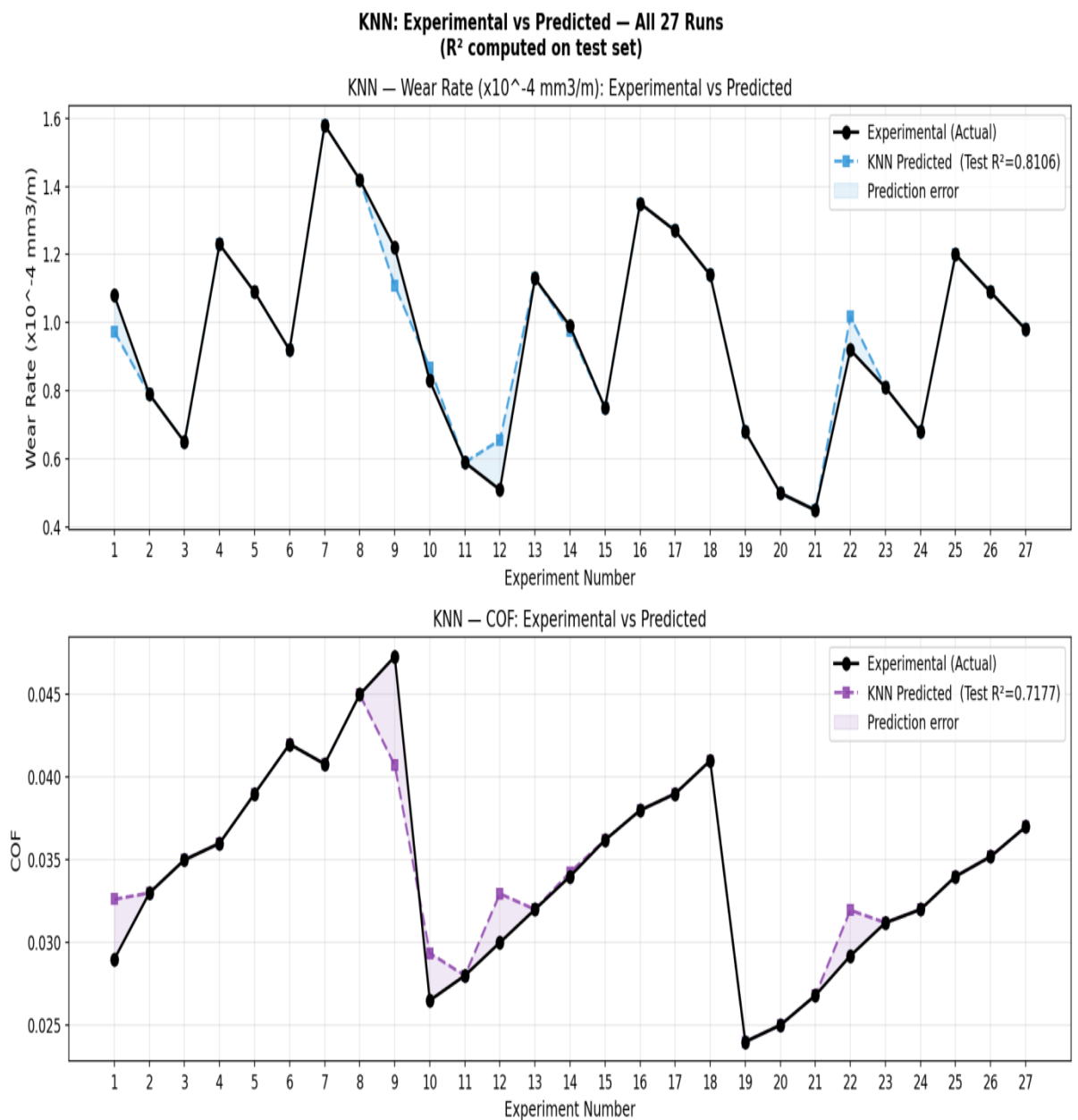


Fig.2.12. KNN Predicted vs. Actual Values

Comparative Analysis of Machine Learning Model :

- The model comparison graph presents the predictive performance of ANN, Random Forest, SVR, and KNN for Wear Rate and COF prediction.
- Performance metrics such as R^2 , RMSE, MAE, and MAPE were used for evaluation . Among all models, the model with the highest R^2 and lowest error values exhibited superior prediction capability.
- The close agreement between predicted and experimental values across all models indicates that machine learning techniques are highly effective for tribological property prediction in Cu-Gr-TiC composites.

MODELS	MEAN ABSOLUTE ERROR (MAE)	ROOT MEAN SQUARE ERROR (RMSE)	COEFFICIENT OF DETERMINATION (R^2)	MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)
ANN (MLP)	0.1056	0.1316	0.6497	11.31
Random Forest	0.0799	0.1004	0.7959	10.06
SVR	0.0592	0.0742	0.8887	6.24
KNN	0.0855	0.0967	0.8106	10.70

Table 1.5. Wear Rate Prediction $\times 10^{(-4)}$ mm³/m units

MODEL	MEAN ABSOLUTE ERROR (MAE)	ROOT MEAN SQUARE ERROR(RMSE)	COEFFICIENT OF DETERMINATION (R^2)	MEAN ABSOLUTE PERCENTAGE ERROR (MAPE)
ANN (MLP)	0.000871	0.001349	0.9619	2.79
Random Forest	0.002805	0.003176	0.7888	8.41
SVR	0.002201	0.002776	0.8387	6.44
KNN	0.003169	0.003671	0.7177	9.53

Table1.6. Coefficient of Friction Prediction

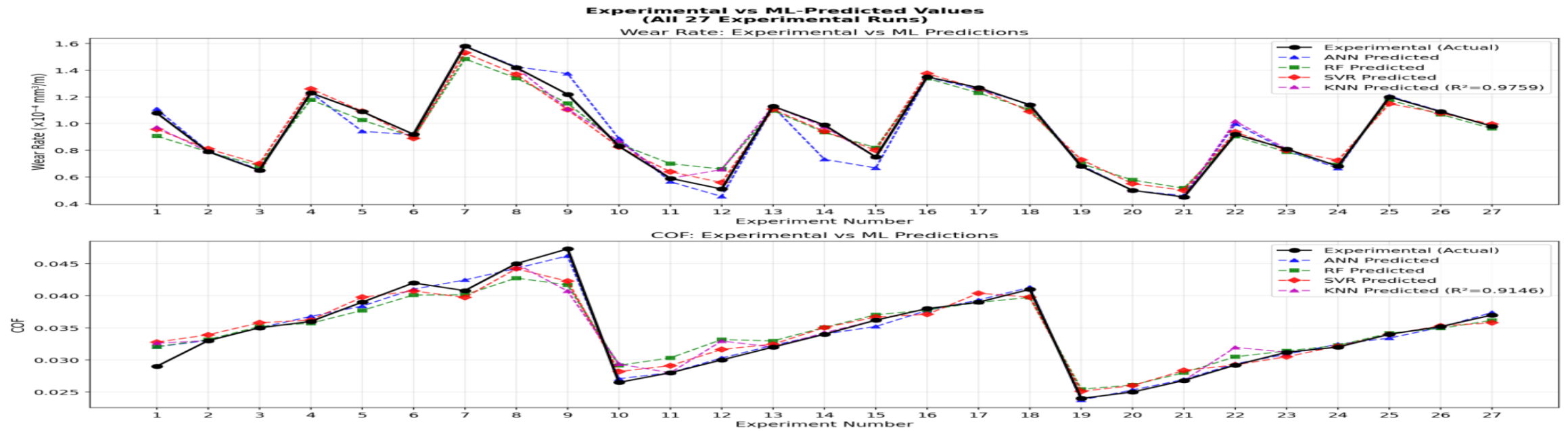


Fig.2.14. Experimental VS Predicted

- Fig. shows how each ML model's predicted values follow the actual experimental values for both wear rate and COF.
- For wear rate, SVR tracked the experimental curve most closely with $R^2 = 0.8887$, staying near the actual values even at sharp peaks and dips.
- For COF, ANN followed the experimental line so precisely that both lines were almost overlapping throughout all runs, achieving $R^2 = 0.9619$. RF and KNN also followed the general trend but missed slightly at extreme values where the curve changed sharply.
- This plot confirms that SVR is the most reliable model for wear rate and ANN is the most reliable model for COF prediction in this study.

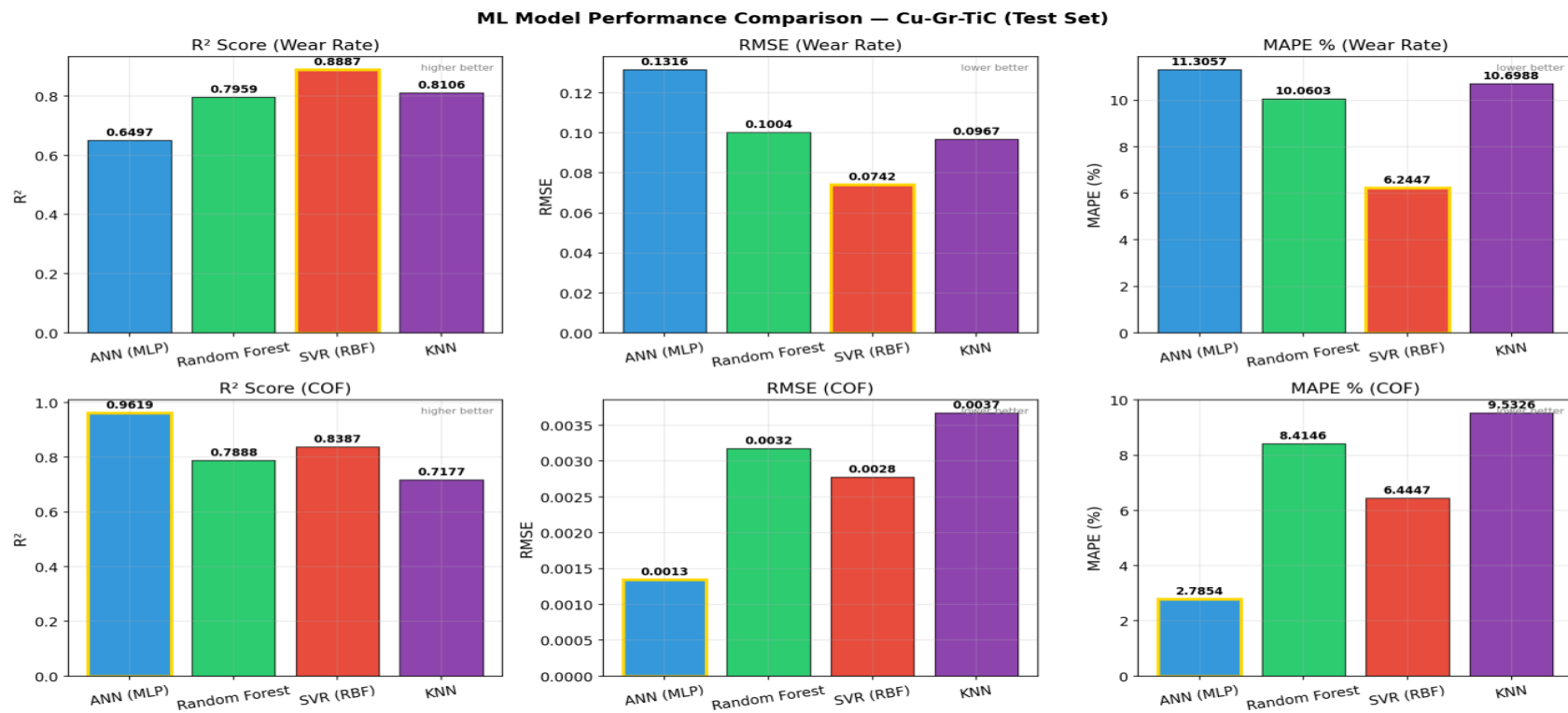


Fig.2.15. ML Model Performance Comparison

- Fig. presents a side by side comparison of all four models using three performance metrics for both wear rate and COF on the test set.
- For wear rate, SVR clearly stands out with the highest R^2 of 0.8887, lowest RMSE of 0.0742, and lowest MAPE of 6.24%, meaning it made the least prediction error among all models. ANN performed the weakest for wear rate with R^2 of only 0.6497 and MAPE of 11.31%, which is likely because the dataset was too small for ANN to train properly. For COF prediction the situation flipped completely, where ANN became the best model with R^2 of 0.9619 and an impressively low MAPE of just 2.79%.
- This figure makes it very clear that the best model is different for each output, and choosing one model for everything would compromise accuracy.

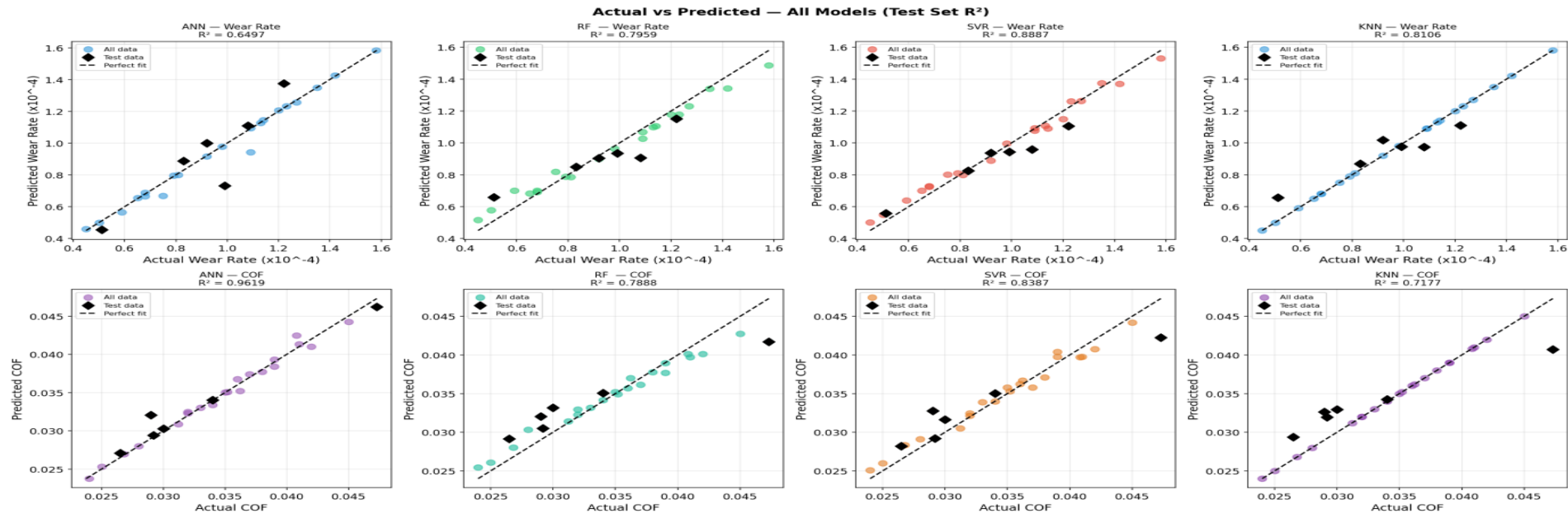


Fig.2.16. Actual Vs Predicted –All Models (Test set R²)

- The fig. plots predicted values against actual experimental values for all four models, where the dashed diagonal line represents a perfect prediction.
- SVR showed the tightest clustering along the diagonal for wear rate ($R^2 = 0.8887$), while ANN produced the cleanest alignment for COF ($R^2 = 0.9619$), with points almost perfectly sitting on the ideal line.
- ANN struggled visibly for wear rate and KNN showed the most scatter for COF, with points drifting away at extreme values. Notably, test set points in all models followed the same pattern as the full dataset, confirming that no model overfitted and all generalized reliably to unseen experimental conditions

Test Result

S.NO	Input Parameter	ANN	SVR	KNN	RF
1.	35,5000,2	1.1091	0.9064	1.0133	0.8071
2.	40,5200,2.4	0.5644	0.5999	0.6267	0.6497
3.	55,7500,4	0.6651	0.6773	0.7833	0.7272

Table 1.6. Prediction Comparison for Wear Rate

S.NO	Input Parameter	ANN	SVR	KNN	RF
1.	35,5000,2	0.032	0.0324	0.0307	0.0327
2.	40,5200,2.4	0.028	0.0293	0.0287	0.0296
3.	55,7500,4	0.0325	0.032	0.0287	0.0326

Table 1.7. Prediction Comparison for COF

Conclusions:

- SVR with RBF kernel achieved the best Wear Rate prediction performance ($R^2 = 0.8887$, MAPE = 6.24%), outperforming Random Forest ($R^2 = 0.7959$), KNN ($R^2 = 0.8106$), and ANN ($R^2 = 0.6497$).
- ANN (MLP, 64-32-16 architecture) achieved exceptional COF prediction accuracy ($R^2 = 0.9619$, MAPE = 2.79%), the highest performance of any model for any output in this study.
- Among all models, SVR demonstrated the most balanced and consistent predictive performance across both output variables, achieving $R^2 = 0.8887$ for Wear Rate and $R^2 = 0.8387$ for COF simultaneously. No other model maintained competitive accuracy for both responses — ANN excelled only for COF while underperforming for Wear Rate, and KNN showed the weakest COF prediction.
- ML-guided optimization identified that minimum Wear Rate ($0.4312 \times 10^{-4} \text{ mm}^3/\text{m}$) and minimum COF (0.0237) are both achieved at Load = 30 N, Distance = 8,000 m, and TiC = 4.5 wt.%, consistent with the physical understanding of the composite system.

References

1. Ankit, Vineet Kumar, Amit Kumar Yadav, Gaurav Gautam, Kamalesh Kumar Singh and Sunil Mohan¹ et.al. “Prediction of tribological performance of Cu-Gr-TiC composites based on response surface methodology and worn surface analysis”, (2023)
2. Huifeng NING, Faqiang CHEN, Yunfeng SU, Hongbin LI, Hengzhong FAN, Junjie SONG, Yongsheng ZHANG, Litian HU et.al. “Modeling and prediction of tribological properties of copper/ aluminum-graphite self-lubricating composites using machine learning algorithms”, (2023)
3. Ankit, Gaurav Gautam, Kamalesh, K Singh and Sunil Mohan et.al, “Synergetic influence of TiCnp and graphite particles on tribological performance of Cu based composites prepared by flake powder metallurgy”, (2023)
4. Pranav Dev Srivyas, M.S. Charoo et.al. “Role Of Reinforcements On The Mechanical And Tribological Behavior Of Aluminum Metal Matrix Composites”, (2018)
5. S.V. Prasad, and R. Asthana et.al. , “Aluminum metal–matrix composites for automotive applications: tribological considerations”, Vol. 17, No. 3, October 2004
6. N.G. Siddeshkumar, H.M. Manjula, K. Balakrishnan, T.C. Pramod, B. Latha Shankar, Arshan Ali Khan, H.M. Pruthvi, Santosh Pawan K, M.B. Nandakumari, Subramanya R. Prabhu, C. Durga Prasad et.al. “Study on dry sliding wear behavior and machine learning models for wear rate prediction of nano hybrid 2219 MMCs reinforced with n-B₄C & MoS₂”, (2025)
7. Barun Haldar, Hillol Joardar, Arpan Kumar Mondal, Nashmi H. Alrasheedi, Rashid Khan and Murugesan P. Papathe et.al. “Machine-Learning-Driven Analysis of Wear Loss and Frictional Behavior in Magnesium Hybrid Composites”, (2025)